

Machine Learning & AI

Graduate course overview · Biotechnology M.S., California State University San Marcos

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Course description

A graduate introduction to machine learning for biotechnologists, spanning data analytics, clustering, classification, deep learning, large language models, and AI ethics. Through lectures, hands-on projects, and case studies, students gain the practical and theoretical grounding to apply machine learning to real problems in the life sciences.

Learning objectives

1. Understand the core principles and algorithms of machine learning.
2. Preprocess, analyze, and visualize data using Python and data-science notebooks.
3. Build, train, and evaluate machine-learning models on biotechnology use cases.
4. Explore deep-learning architectures and their applications.
5. Understand the development and implications of large language models.
6. Address the ethical considerations and societal impacts of AI.

Curriculum at a glance

- 01 Foundations of data science** Types of learning, the evolution of ML, and the languages, tools, and notebooks used in practice.
- 02 Statistics & exploratory data analysis** Descriptive and inferential statistics, data collection and cleaning, sampling, and guarding against bias.
- 03 Supervised learning** Regression and classification with logistic regression, decision trees, random forests, SVMs, and KNN.
- 04 Model evaluation & tuning** Train/test splits, cross-validation, hyperparameter tuning, and metrics such as accuracy, F1, and ROC.
- 05 Unsupervised learning** Clustering with K-means and hierarchical methods, and where it applies in the life sciences.
- 06 Deep learning** Neural-network fundamentals: perceptrons, backpropagation, activation functions, CNNs, and RNNs.
- 07 Generative AI & agents** Generative models, agentic architectures, and applications such as AI agents in drug discovery.
- 08 Large language models** Transformers, attention, tokenization, and how modern LLMs represent and generate language.

Applied biotechnology case studies

- Breast-cancer data exploration
- Gene-expression modeling for cancer drug response
- Heart-disease classification models built from scratch

Ethics throughout

Ethics is woven through the course rather than saved for the end. Students examine the societal impact of AI — algorithmic bias and fairness, surveillance and privacy, and the dual-use risks that are especially acute in biotechnology, where the same tools that accelerate discovery can also enable harm. The course connects these questions to current work on AI safety and responsible deployment.

Assessment

Quizzes	5%
Assignments	10%
Midterm exam	30%
Presentations & projects	25%
Final exam	30%